

JSS 2 MATHEMATICS – FIRST TERM

TOPIC 9: ALGEBRAIC EXPRESSIONS II

1. Expansion of Algebraic Expressions

Expansion means removing brackets by multiplying. Example: $3(x + 4) = 3x + 12$. For two brackets: $(x + 2)(x + 3) = x^2 + 5x + 6$ (use FOIL or distributive law).

2. Factorization of Quadratic Expressions

A quadratic expression has the form $ax^2 + bx + c$. Factorisation seeks two binomials whose product is the quadratic. Example: $x^2 + 5x + 6 = (x + 2)(x + 3)$.

3. Algebraic Fractions

An algebraic fraction has a variable in the numerator or denominator (or both), e.g. $3/x$, $(x+1)/4$, $2a/b$.

4. Addition and Subtraction of Algebraic Fractions (Whole Number Denominators)

Find a common denominator (often the LCM of the denominators), then add or subtract the numerators.

Example: $1/2 + 1/3 = 3/6 + 2/6 = 5/6$

Example: $x/2 + x/4 = 2x/4 + x/4 = 3x/4$

5. Application and Quantitative Reasoning

Algebraic fractions appear in problems involving rates, sharing and formulas. Practise simplifying and combining them carefully.

6. Summary

Expand brackets by multiplication; factorise quadratics into two brackets when possible. Algebraic fractions are added/subtracted using a common denominator. Accurate work with signs and common factors is essential.

End of Topic 9 – Algebraic Expressions II

— End of JSS 2 Mathematics First Term Notes —